Speech Signal Processing
Module-3: Automatic Speech Recognition (ASR)

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Outline

1. Introduction
2. Feature extraction
3. Acoustic model

ASR equation

$$\hat{W} = \text{argmax}\{p\left(W|O\right)\}$$

ASR equation

$$\hat{W} = \text{argmax}\{p\,(W|O)\}$$

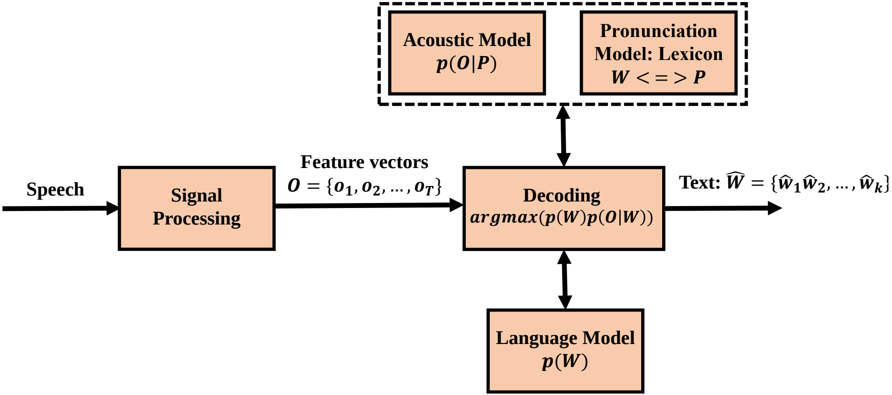
$$= \text{argmax}\{p(W)p\,(O|W)\}$$

ASR equation

$$\hat{W} = \text{argmax}\{p(W|O)\}$$
$$= \text{argmax}\{p(W)p(O|W)\}$$

1. What data is needed for modelling $p(W)$ and $p(O|W)$?

ASR building blocks



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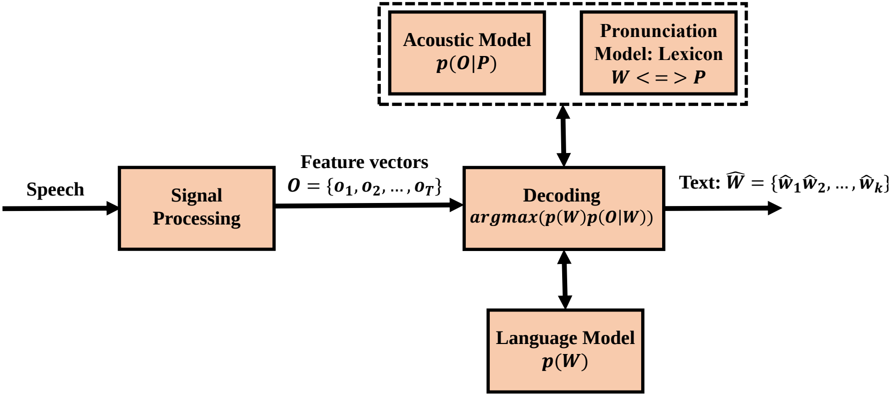

1 Why modelling $p(O|P)$ is efficient?

Word and phoneme sequence mapping: Lexicon?

An exemplary Lexicon

```
BRANER  B R EY1 N ER0
BRANFORD  B R AE1 N F ER0 D
BRANHAM  B R AE1 N HH AH0 M
BRANI  B R AE1 N IY0
BRANIFF  B R AE1 N IH0 F
BRANIFF'S  B R AE1 N IH0 F S
BRANIGAN  B R AE1 N IH0 G AH0 N
BRANILLO  B R AH0 N IH1 L OW0
BRANIN  B R AE1 N IH0 N
BRANISLOV  B R AE1 N IH0 S L AA2 V
BRANITZKY  B R AH0 N IH1 T S K IY1
BRANK  B R AE1 NG K
BRANK'S  B R AE1 NG K S
BRANKI  B R AE1 NG K IY0
BRANKO  B R AE1 NG K OW0
BRANKS  B R AE1 NG K S
BRANN  B R AE1 N
BRANNA  B R AE1 N AH0
BRANNAM  B R AE1 N AH0 M
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1. Knowing $p(O|P)$ sufficient for computing $p(O|W)$?

Word and phoneme sequence mapping: Lexicon?

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```
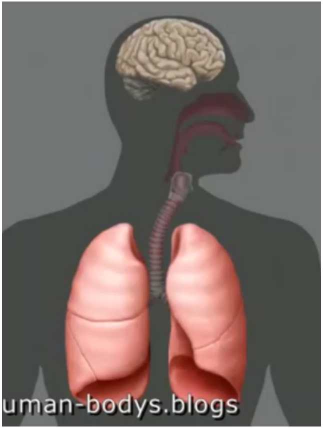

1. Knowing $p(O|P)$ sufficient for computing $p(O|W)$?
2. Knowing W sufficient for obtaining P?

Take-away

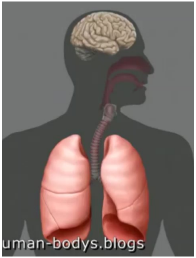
1. ASR Goal: Predict W from O using LM: $p(W)$ and AM: $p(O|P)$.
2. O is encode the relevant information representing speech acoustics at the frame level.

Speech production

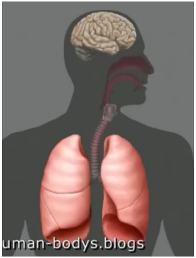


1. What is the behaviour of the speech production system?

Quasi stationary behaviour

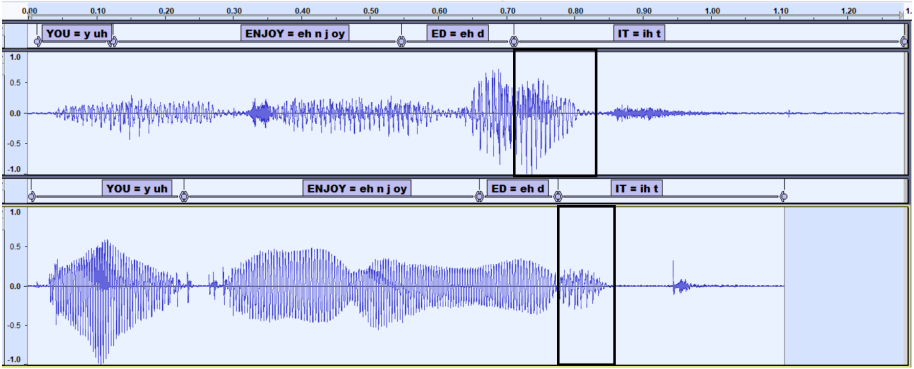


Quasi stationary behaviour



1. Any stationary behaviour can be observed?
2. At what level, that is stationary?

Non-stationary, but there is a structure in small segment



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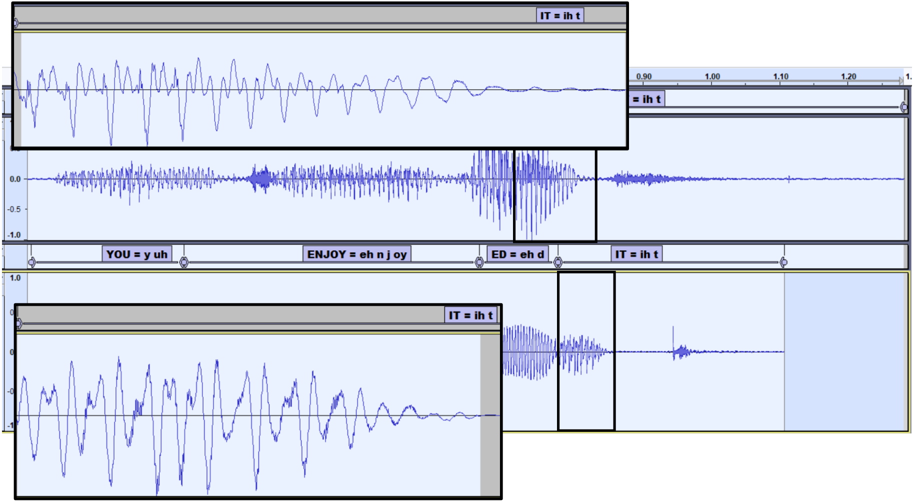

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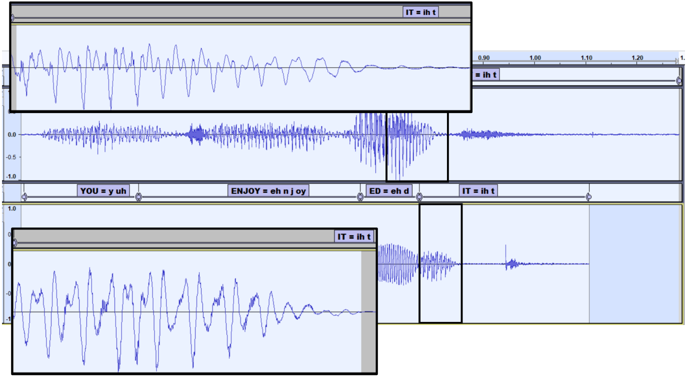

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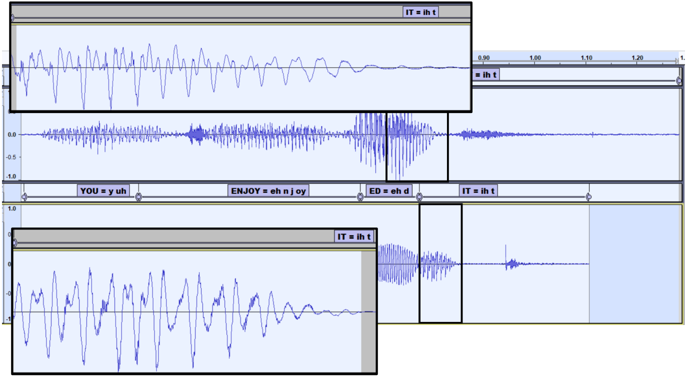
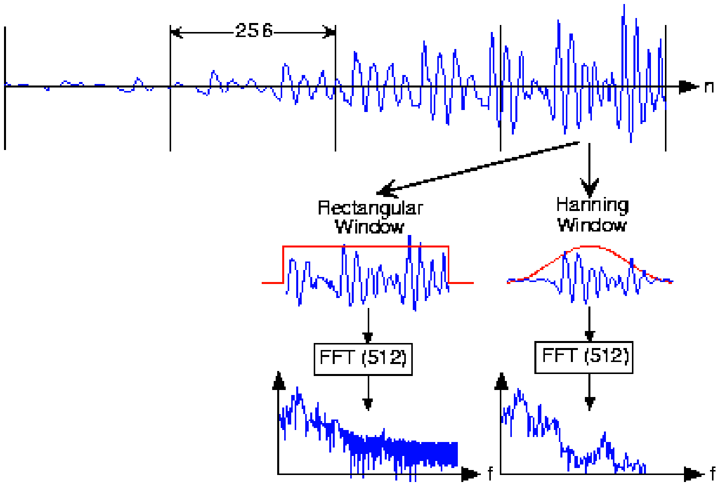
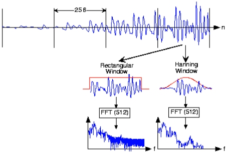


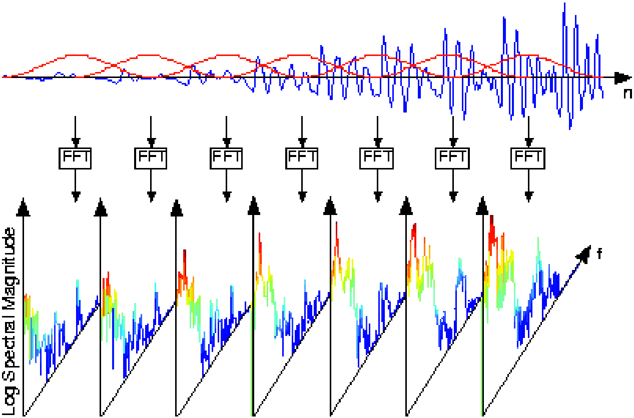
1. What is the difference between frames and segments?

Frame based analysis

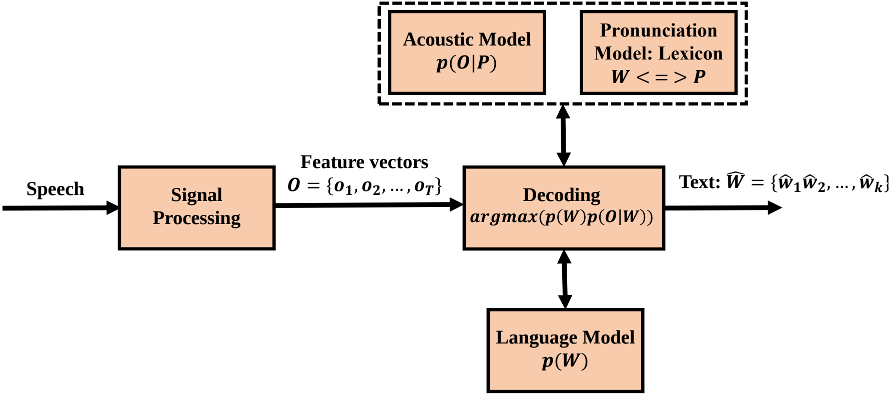


Frame based analysis

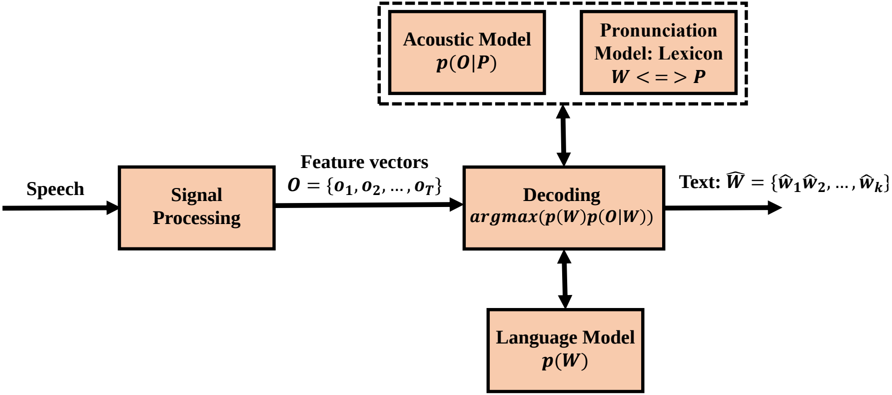




ASR building blocks



ASR building blocks



1. What are the relative sizes of O, P and W?

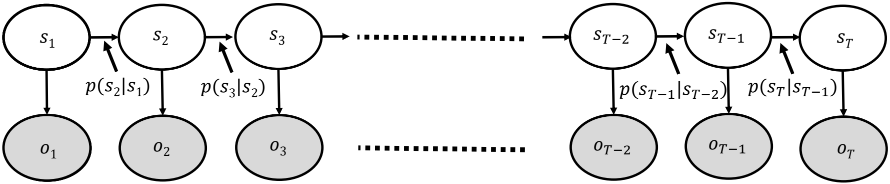
Take-away

1. ASR Goal: Predict W from O using LM: $p(W)$ and AM: $p(O|P)$.
2. O is encode the relevant information representing speech acoustics at the frame level.

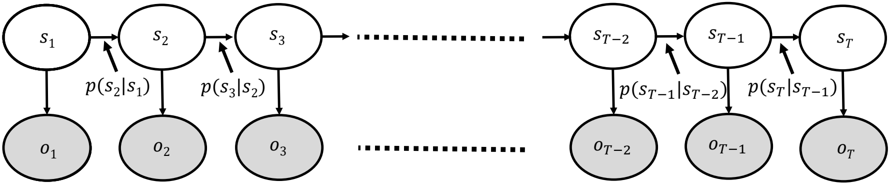
Take-away

1. ASR Goal: Predict W from O using LM: $p(W)$ and AM: $p(O|P)$.
2. O is encode the relevant information representing speech acoustics at the frame level.
3. Speech production system is time-varying, but it exhibits **stationary behaviour** at the segment level, mainly at **phoneme sub-segmental** level.
4. However, the duration of the phoneme sub-segments are **dynamic**.
5. The stationary behaviour of the phoneme sub-segment is due to the fixed vocal tract shape, whose information is **hidden** in the speech acoustics (O).

Hidden Markov Model (HMM)

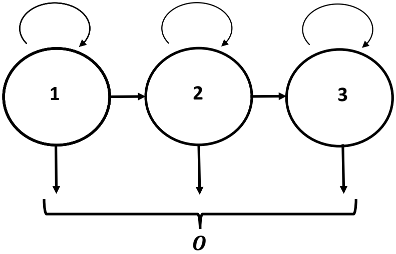


Hidden Markov Model (HMM)

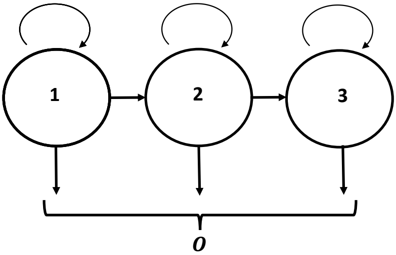


1. What does the hidden state model?
2. What is need of Markov chain?

Left-to-right three state HMM

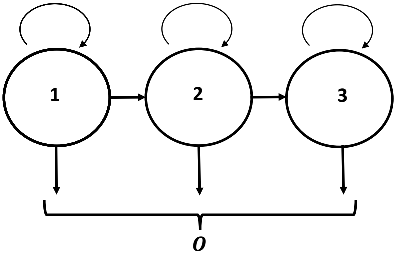


Left-to-right three state HMM



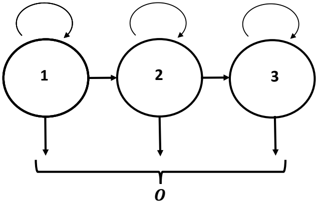
1. Why three state?

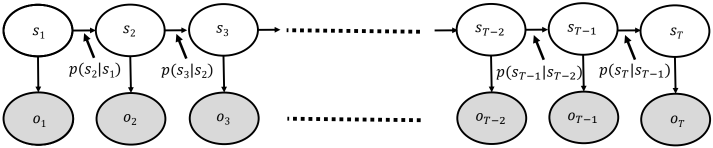
Left-to-right three state HMM



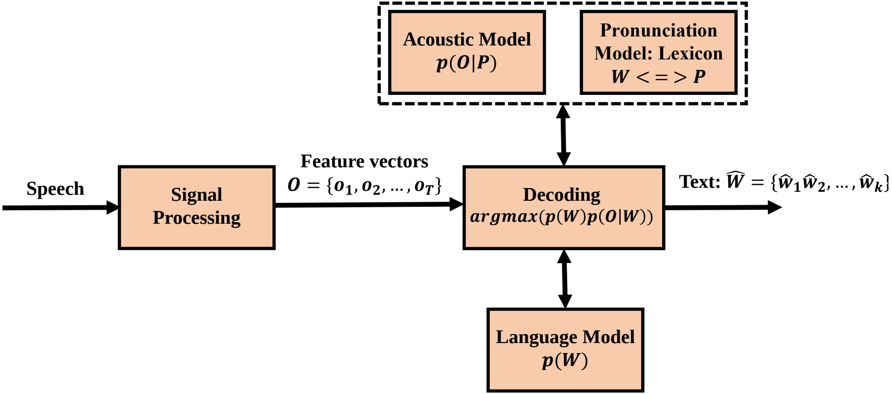
1. Why three state?
2. Why left-to-right is sufficient for AM?

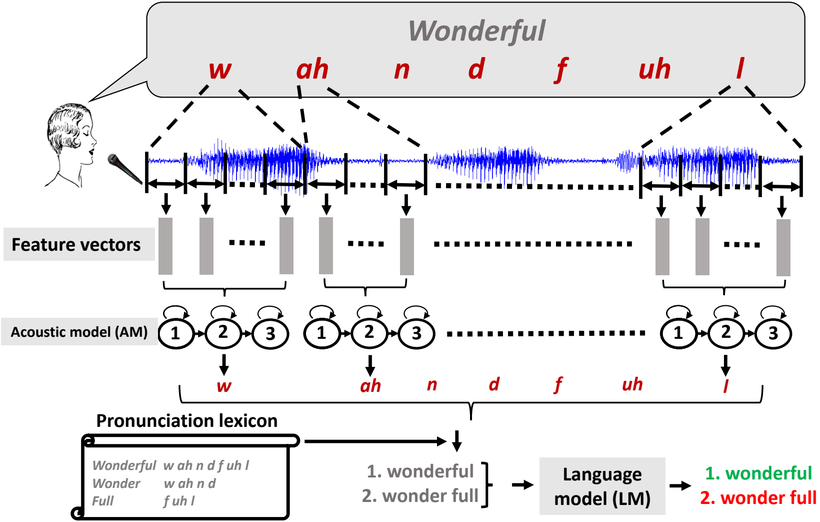
Left-to-right three state HMM



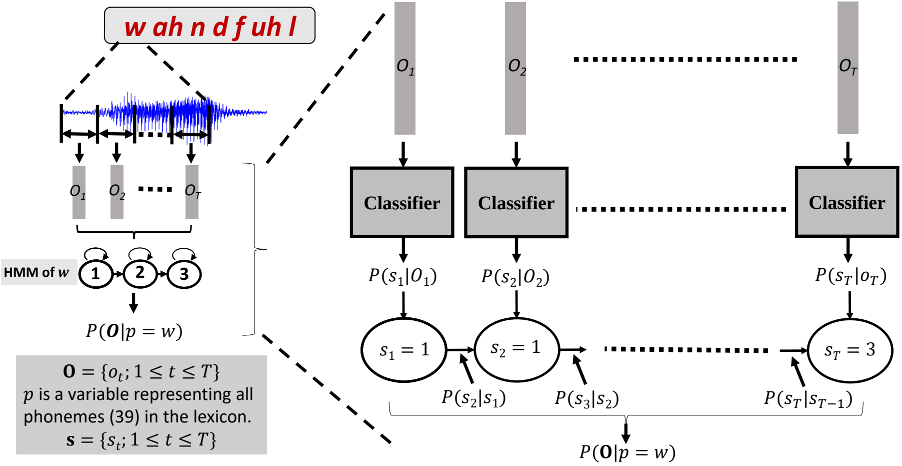


ASR building blocks



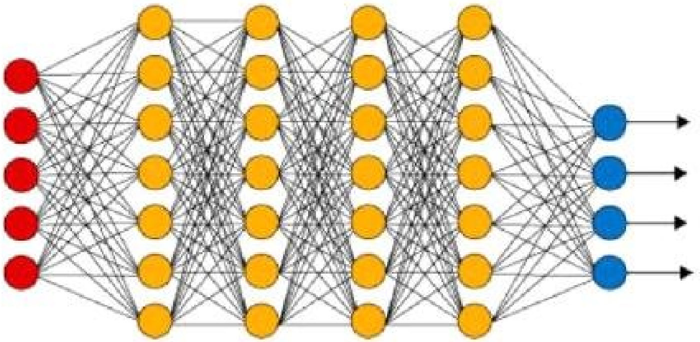

Wonderful
w ah n d f uh l
Feature vectors
Acoustic model (AM)
1 2 3 1 2 3 1 2 3
w ah n d f uh l
Pronunciation lexicon
Wonderful w ah n d f uh l
Wonder w ah n d
Full f uh l
1. wonderful
2. wonder full
Language model (LM)
1. wonderful
2. wonder full

Phoneme likelihood computation

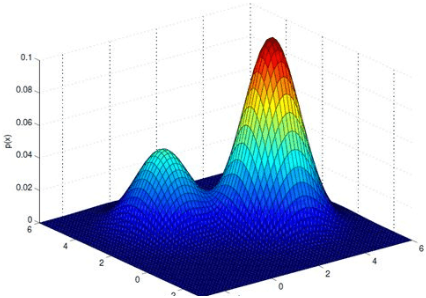


$\mathbf{O} = \{o_t; 1 \le t \le T\}$
p is a variable representing all phonemes (39) in the lexicon.
$\mathbf{s} = \{s_t; 1 \le t \le T\}$

DNN-HMM



GMM-HMM



Thank you